

T.ASHWINI

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G3-(33)

Data Analytics Assignment

Data Cleaning

```
In [32]: import pandas as pd
file_path = r"C:\Users\Aswini\Downloads\Atal_Jal_Area_0.csv"
df = pd.read_csv(file_path)
df
```

```
Out[32]:
```

	Sr. No.	State_Name	District_Name	Block_Name	Gram_Panchayat_Name
0	1	Gujarat_24	Banaskantha_441	Dantiwada_3735	Akoli_155943
1	2	Gujarat_24	Banaskantha_441	Dantiwada_3735	Arkhi_155944
2	3	Gujarat_24	Banaskantha_441	Dantiwada_3735	Bhadali (Zat)_155945
3	4	Gujarat_24	Banaskantha_441	Dantiwada_3735	Bhadli Kotha_155946
4	5	Gujarat_24	Banaskantha_441	Dantiwada_3735	Bhakhar Moti_155947
...	...	...	...	...	...
8215	8216	Uttar Pradesh_9	Shamli_660	Kandhala_1430	Rampur Kheri_81319
8216	8217	Uttar Pradesh_9	Shamli_660	Kandhala_1430	Rasoolpur Gujran_81320
8217	8218	Uttar Pradesh_9	Shamli_660	Kandhala_1430	Salffa_271769
8218	8219	Uttar Pradesh_9	Shamli_660	Kandhala_1430	Sunnaa_271752
8219	8220	Uttar Pradesh_9	Shamli_660	Kandhala_1430	Taharpur Bhabisa_271764

8220 rows × 5 columns

```
In [34]: df.info
```

```
Out[34]: <bound method DataFrame.info of
Block_Name \
0      1      Gujarat_24 Banaskantha_441 Dantiwada_3735
1      2      Gujarat_24 Banaskantha_441 Dantiwada_3735
2      3      Gujarat_24 Banaskantha_441 Dantiwada_3735
3      4      Gujarat_24 Banaskantha_441 Dantiwada_3735
4      5      Gujarat_24 Banaskantha_441 Dantiwada_3735
...      ...      ...
8215    8216 Uttar Pradesh_9 Shamli_660 Kandhala_1430
8216    8217 Uttar Pradesh_9 Shamli_660 Kandhala_1430
8217    8218 Uttar Pradesh_9 Shamli_660 Kandhala_1430
8218    8219 Uttar Pradesh_9 Shamli_660 Kandhala_1430
8219    8220 Uttar Pradesh_9 Shamli_660 Kandhala_1430

Gram_Panchayat_Name
0      Akoli_155943
1      Arkhi_155944
2      Bhadali (Zat)_155945
3      Bhadli Kotha_155946
4      Bhakhar Moti_155947
...      ...
8215    Rampur Kheri_81319
8216    Rasoolpur Gujran_81320
8217    Salffa_271769
8218    Sunnaa_271752
8219    Taharpur Bhabisa_271764

[8220 rows x 5 columns]>
```

```
In [30]: print(df.head())
```

	Sr. No.	State_Name	District_Name	Block_Name	Gram_Panchayat_Name
0	1	Gujarat_24	Banaskantha_441	Dantiwada_3735	Akoli_155943
1	2	Gujarat_24	Banaskantha_441	Dantiwada_3735	Arkhi_155944
2	3	Gujarat_24	Banaskantha_441	Dantiwada_3735	Bhadali (Zat)_155945
3	4	Gujarat_24	Banaskantha_441	Dantiwada_3735	Bhadli Kotha_155946
4	5	Gujarat_24	Banaskantha_441	Dantiwada_3735	Bhakhar Moti_155947

```
In [28]: df.tail()
```

```
Out[28]:
```

	Sr. No.	State_Name	District_Name	Block_Name	Gram_Panchayat_Name
8215	8216	Uttar Pradesh_9	Shamli_660	Kandhala_1430	Rampur Kheri_81319
8216	8217	Uttar Pradesh_9	Shamli_660	Kandhala_1430	Rasoolpur Gujran_81320
8217	8218	Uttar Pradesh_9	Shamli_660	Kandhala_1430	Salffa_271769
8218	8219	Uttar Pradesh_9	Shamli_660	Kandhala_1430	Sunnaa_271752
8219	8220	Uttar Pradesh_9	Shamli_660	Kandhala_1430	Taharpur Bhabisa_271764

```
In [36]: df.drop(columns=["Sr. No."], inplace=True)
```

```
In [38]: df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_")
```

```
In [40]: df = df.applymap(lambda x: x.rsplit('_', 1)[0] if isinstance(x, str) and '_' in x else x)
```

C:\Users\Aswini\AppData\Local\Temp\ipykernel_16832\686454906.py:1: FutureWarning: DataFrame.applymap has been deprecated. Use DataFrame.map instead.

```
df = df.applymap(lambda x: x.rsplit('_', 1)[0] if isinstance(x, str) and '_' in x else x)
```

```
In [42]: df.drop_duplicates(inplace=True)
```

```
In [44]: df = df.applymap(lambda x: x.strip() if isinstance(x, str) else x)
```

C:\Users\Aswini\AppData\Local\Temp\ipykernel_16832\242626680.py:1: FutureWarning: DataFrame.applymap has been deprecated. Use DataFrame.map instead.

```
df = df.applymap(lambda x: x.strip() if isinstance(x, str) else x)
```

```
In [46]: df.info()
print(df.head())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8220 entries, 0 to 8219
Data columns (total 4 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   state_name            8220 non-null   object
1   district_name         8220 non-null   object
2   block_name            8220 non-null   object
3   gram_panchayat_name   8220 non-null   object
dtypes: object(4)
memory usage: 257.0+ KB
```

	state_name	district_name	block_name	gram_panchayat_name
0	Gujarat	Banaskantha	Dantiwada	Akoli
1	Gujarat	Banaskantha	Dantiwada	Arkhi
2	Gujarat	Banaskantha	Dantiwada	Bhadali (Zat)
3	Gujarat	Banaskantha	Dantiwada	Bhadli Kotha
4	Gujarat	Banaskantha	Dantiwada	Bhakhar Moti

```
In [48]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8220 entries, 0 to 8219
Data columns (total 4 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   state_name            8220 non-null   object
1   district_name         8220 non-null   object
2   block_name            8220 non-null   object
3   gram_panchayat_name   8220 non-null   object
dtypes: object(4)
memory usage: 257.0+ KB
```

```
In [54]: df.isnull().sum()
```

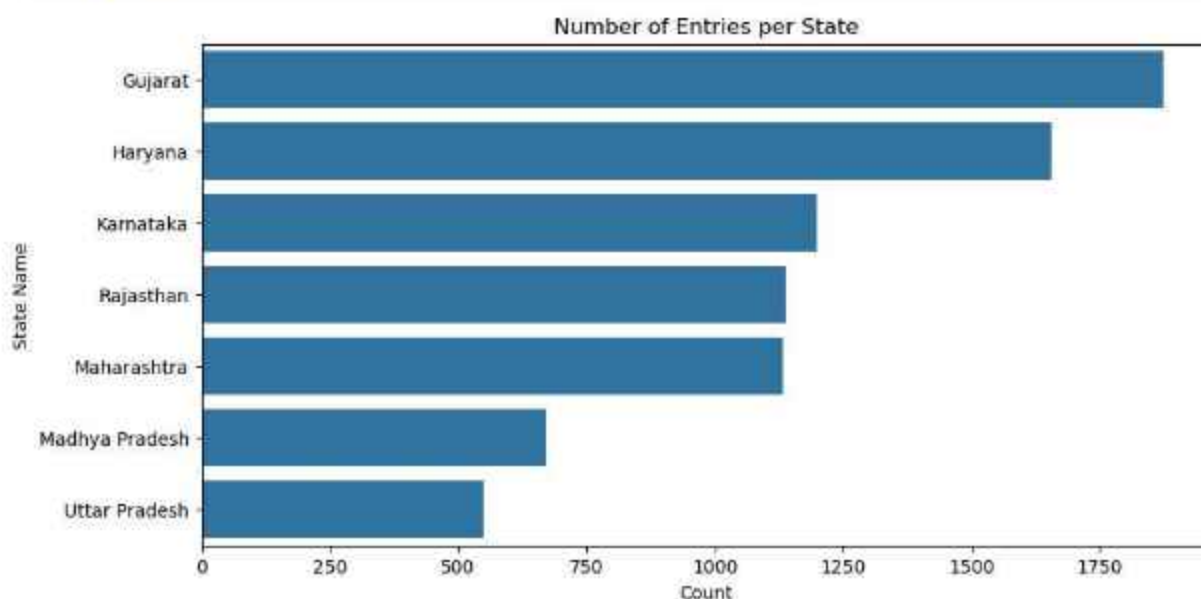
```
Out[54]: state_name      0
district_name    0
block_name       0
gram_panchayat_name  0
dtype: int64
```

Data Visualization

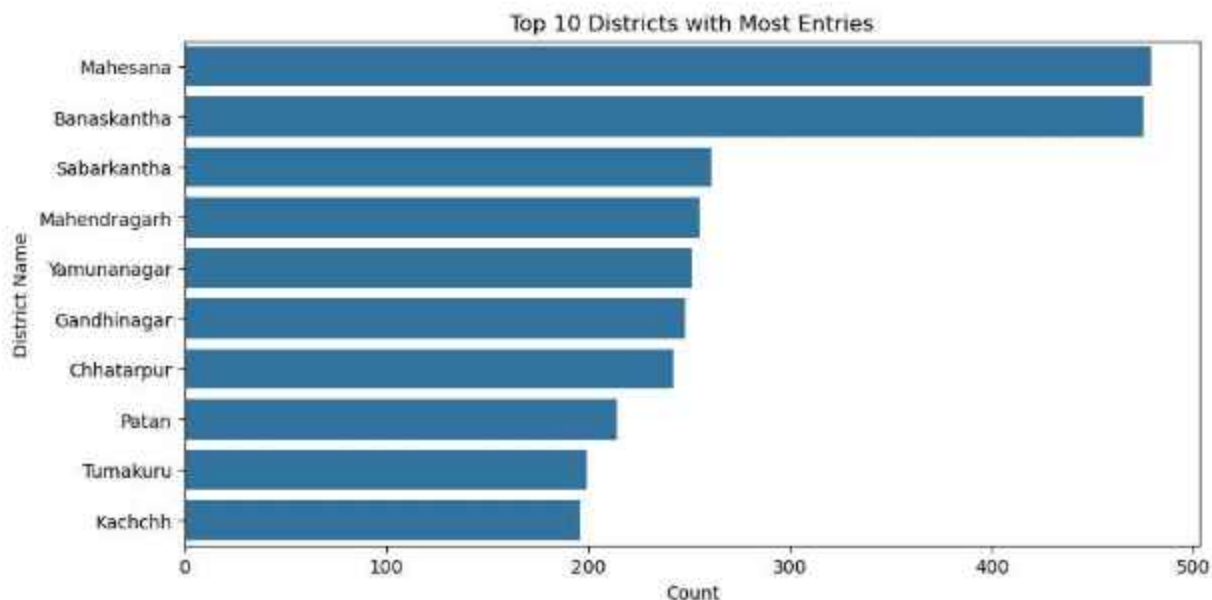
```
In [56]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [60]: file_path = r"C:\Users\Aswini\Downloads\Atal_Jal_Area_0.csv"
df = pd.read_csv(file_path)
```

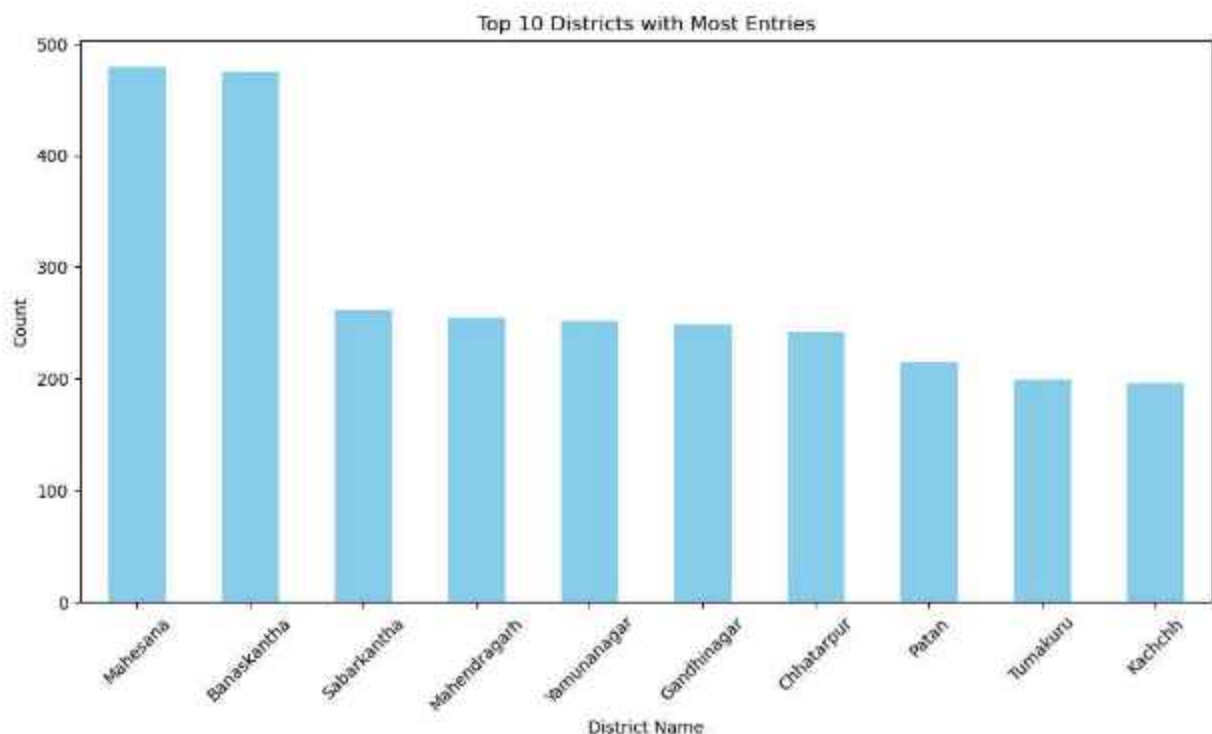
```
In [74]: plt.figure(figsize=(10, 5))
sns.countplot(y=df["state_name"], order=df["state_name"].value_counts().index)
plt.title("Number of Entries per State")
plt.xlabel("Count")
plt.ylabel("State Name")
plt.show()
```



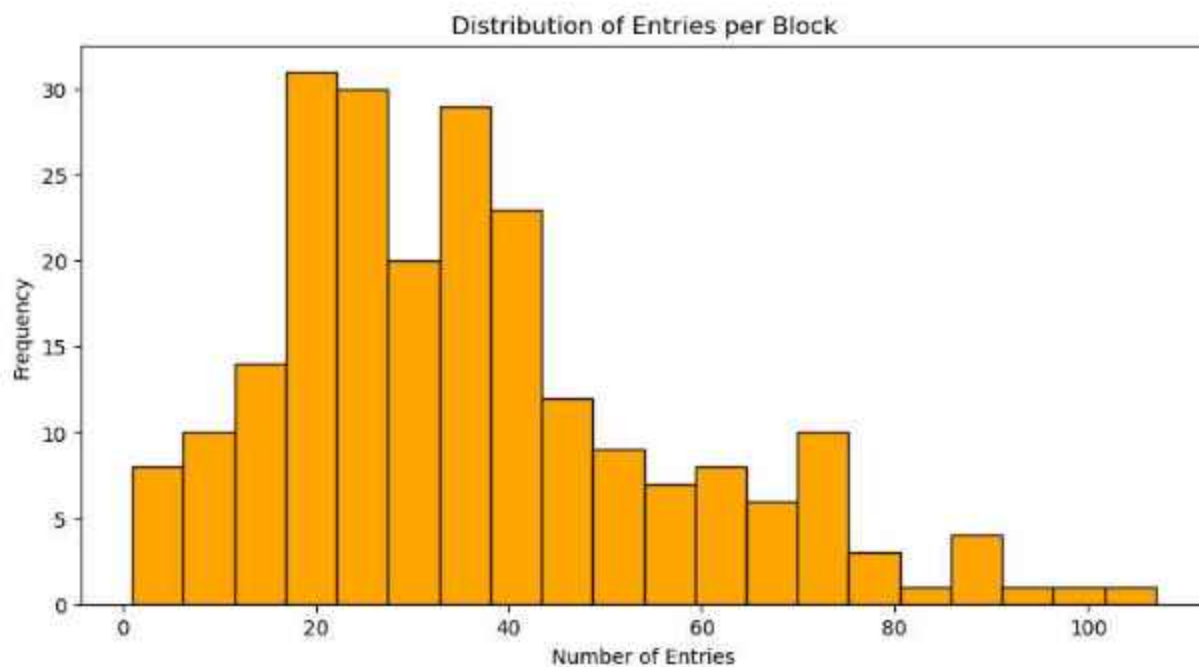
```
In [76]: plt.figure(figsize=(10, 5))
sns.countplot(y=df["district_name"], order=df["district_name"].value_counts().index)
plt.title("Top 10 Districts with Most Entries")
plt.xlabel("Count")
plt.ylabel("District Name")
plt.show()
```

```
In [78]: plt.figure(figsize=(12, 6))
df["district_name"].value_counts().head(10).plot(kind="bar", color="skyblue")
plt.title("Top 10 Districts with Most Entries")
plt.xlabel("District Name")
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.show()
```

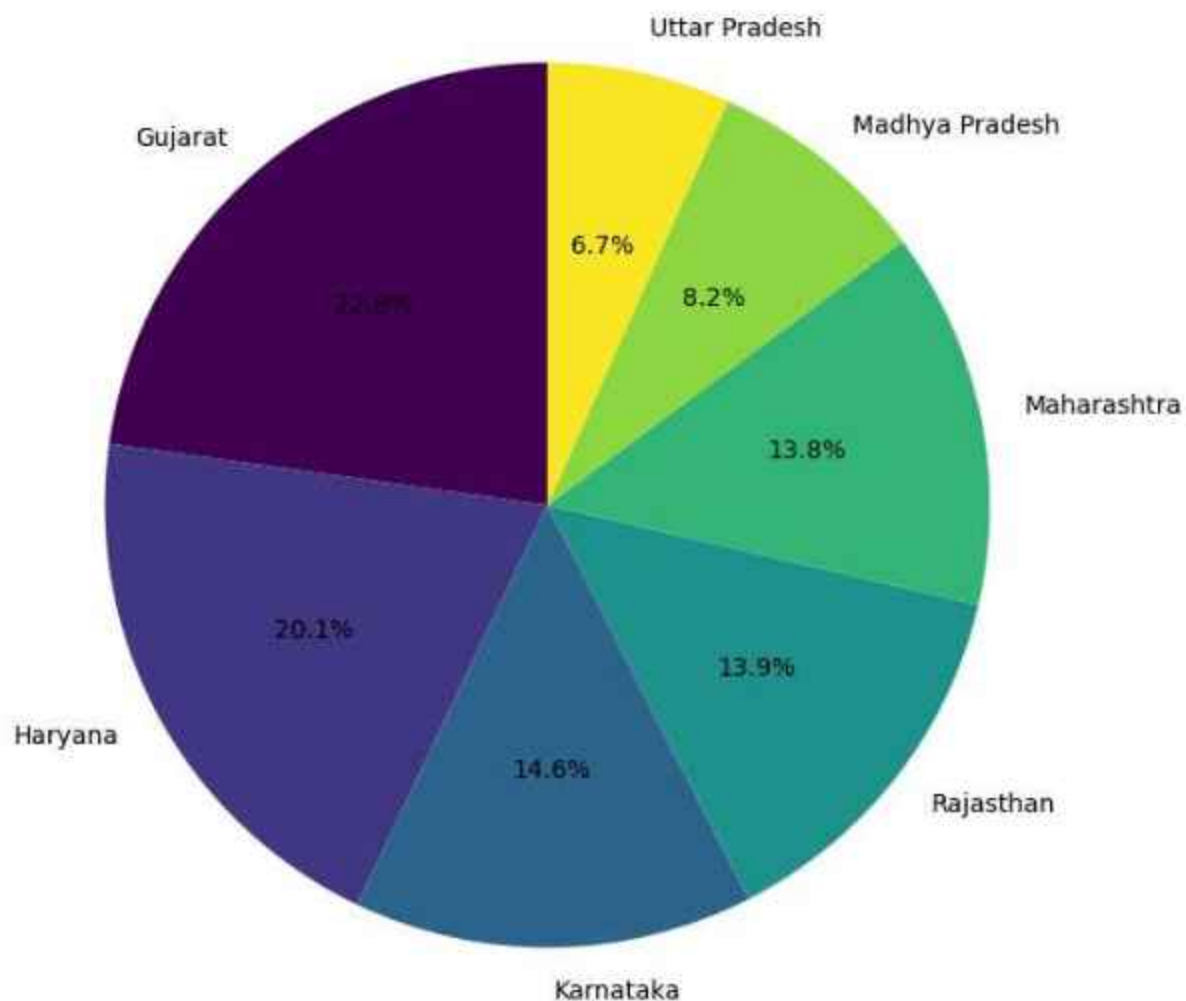


```
In [80]: plt.figure(figsize=(10, 5))
df["block_name"].value_counts().plot(kind="hist", bins=20, color="orange", edgecolor="black")
plt.title("Distribution of Entries per Block")
plt.xlabel("Number of Entries")
plt.ylabel("Frequency")
plt.show()
```

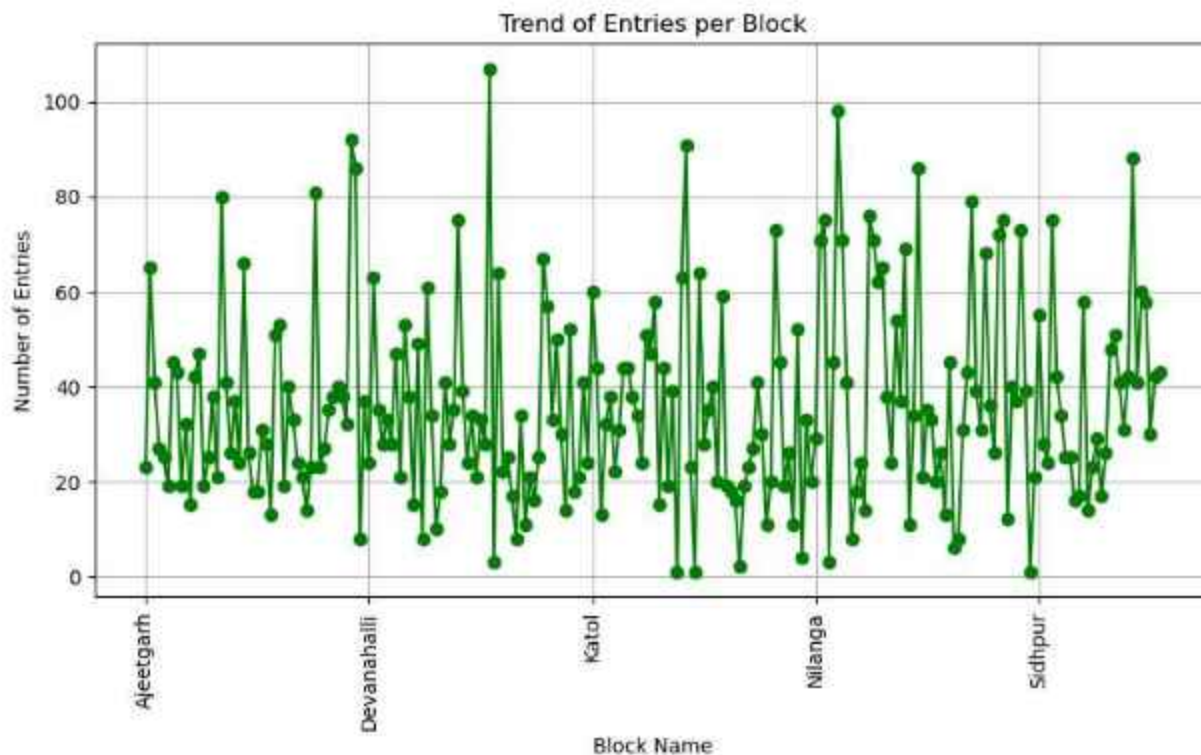


```
In [82]: plt.figure(figsize=(8, 8))
df["state_name"].value_counts().plot(kind="pie", autopct='%1.1f%%', startangle=90,
plt.title("State-wise Distribution of Entries")
plt.ylabel("")
plt.show()
```

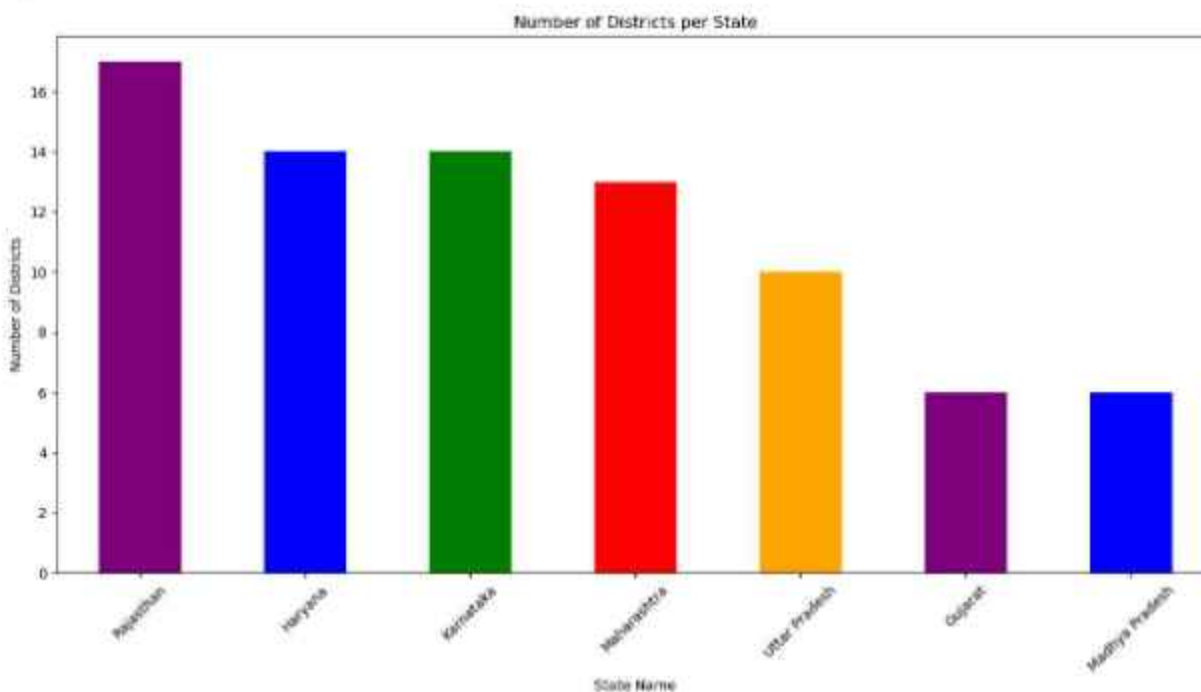
State-wise Distribution of Entries



```
In [84]: plt.figure(figsize=(10, 5))
df["block_name"].value_counts().sort_index().plot(kind="line", marker="o", color="g")
plt.title("Trend of Entries per Block")
plt.xlabel("Block Name")
plt.ylabel("Number of Entries")
plt.xticks(rotation=90)
plt.grid()
plt.show()
```



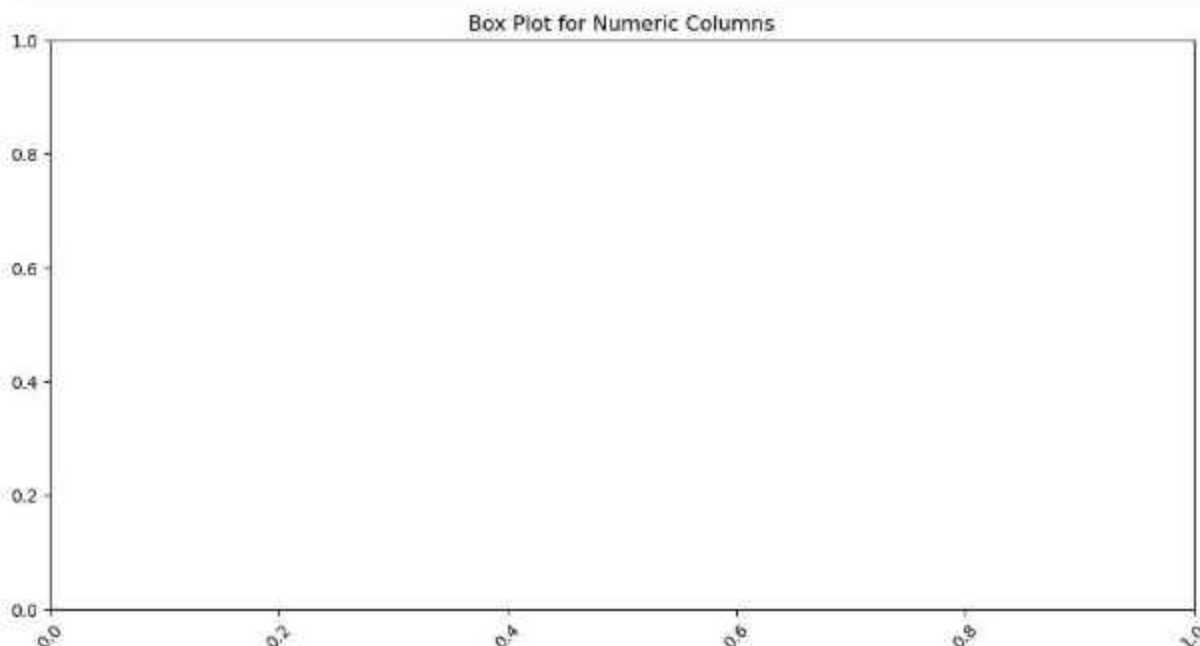
```
In [86]: plt.figure(figsize=(15, 7))
district_counts = df.groupby("state_name")["district_name"].unique().sort_values(a
district_counts.plot(kind="bar", color=["purple", "blue", "green", "red", "orange"])
plt.title("Number of Districts per State")
plt.xlabel("State Name")
plt.ylabel("Number of Districts")
plt.xticks(rotation=45)
plt.show()
```



```
In [88]: plt.figure(figsize=(12, 6))
sns.boxplot(data=df.select_dtypes(include=['number']))
```



```
plt.title("Box Plot for Numeric Columns")
plt.xticks(rotation=45)
plt.show()
```



```
In [90]: plt.figure(figsize=(10, 5))
if df.select_dtypes(include=['number']).shape[1] >= 2:
    numeric_cols = df.select_dtypes(include=['number']).columns[:2]
    sns.lmplot(x=numeric_cols[0], y=numeric_cols[1], data=df, aspect=1.5, scatter_k
    plt.title("Linear Relationship between {} and {}".format(numeric_cols[0], numer
else:
    print("Not enough numeric columns for a linear plot.")
plt.show()
```

Not enough numeric columns for a linear plot.
<Figure size 1000x500 with 0 Axes>

```
In [92]: plt.figure(figsize=(10, 5))
if df.select_dtypes(include=['number']).shape[1] >= 2:
    numeric_cols = df.select_dtypes(include=['number']).columns[:2]
    plt.scatter(df[numeric_cols[0]], df[numeric_cols[1]], alpha=0.5, color='blue')
    plt.title("Scatter Plot between {} and {}".format(numeric_cols[0], numeric_cols
    plt.xlabel(numeric_cols[0])
    plt.ylabel(numeric_cols[1])
    plt.grid()
else:
    print("Not enough numeric columns for a scatter plot.")
plt.show()
```

Not enough numeric columns for a scatter plot.
<Figure size 1000x500 with 0 Axes>

In []: